

SHAALE-VIKAS

School-Alumni Bridge | Empowering Rural Education Infrastructure

Android App Development | GenAI-Powered | Education & Infrastructure Domain

1. Executive Summary

Shaale-Vikas (Kannada: 'School Development') is a mobile-first, Android application that acts as a transparent bridge between rural government schools and their alumni networks. The app empowers school headmasters to publish granular infrastructure needs (micro-needs) while enabling alumni — many of whom have moved to cities — to view, pledge support, and track the real-time impact of their contributions. Powered by Firebase and enriched with GenAI capabilities, Shaale-Vikas transforms goodwill into accountable, measurable action.

2. Problem Statement

Rural government schools across India face chronic infrastructure deficits — leaking roofs, broken furniture, absent sanitation, and fading classrooms. The traditional funding pipeline is slow: needs are reported, escalated through bureaucratic layers, and funds arrive months or years later — by which time the damage has multiplied.

Core Pain Points

- **Delayed Government Funding:** By the time repair funds are sanctioned, minor issues escalate into major structural damage.
- **Invisible Alumni Goodwill:** Thousands of school alumni genuinely want to contribute but lack a reliable, transparent channel to know what is needed.
- **Lack of Transparency:** Donors have no visibility into how their money or goods are used, eroding trust.
- **No Prioritisation Mechanism:** Schools cannot signal urgency or rank their needs to potential supporters.
- **Zero Impact Proof:** There is no 'before and after' record that demonstrates the value of a contribution.

3. Vision & Objectives

Shaale-Vikas envisions a future where every alumni is a stakeholder in their alma mater's growth. The platform is built on three pillars: Community Ownership, Radical Transparency, and Measurable Impact.

Strategic Objectives

- Enable headmasters to publish, manage, and prioritise school micro-needs in real time.
 - Allow alumni to view needs, make simulated pledges, and track funding progress.
 - Build a permanent, visual record of school improvement through impact photos.
 - Create a Donor Hall of Fame to recognise contributors and incentivise participation.
 - Use GenAI to assist headmasters in writing clear need descriptions and cost estimates.
 - Ensure the platform is lightweight, offline-tolerant, and accessible on entry-level Android devices.
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4. Scope of the Application

4.1 In Scope

- Android mobile application (minimum SDK: Android 8.0 / API 26).
- Two user roles: Admin (Headmaster / School Representative) and Alumni (Donor/Pledger).
- Needs Dashboard — create, edit, categorise, and publish school needs with photo and cost estimate.
- Pledge Module — simulated commitment workflow (no real payment gateway).
- Real-time progress tracking via Firebase Firestore.
- Impact Gallery — before/after photo upload and display.
- Donor Hall of Fame — recognition leaderboard for top pledgers.
- GenAI Assistant — AI-powered suggestions for need descriptions and cost estimates.
- Push notifications for pledge milestones and new needs.
- Admin-only authentication and content management controls.

4.2 Out of Scope

- Real money transactions, payment gateway integration, or UPI/banking APIs.
 - iOS, web, or cross-platform versions (Phase 1 is Android-only).
 - Government portal integration or official fund disbursement workflows.
 - Multi-school management or district-level administration.
 - Offline-first synchronisation (basic offline tolerance only).
 - Multi-language localisation beyond English and Kannada.
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5. User Roles & Personas

Role	Persona	Primary Goal
Admin (Headmaster)	School principal / designated teacher managing the school's profile.	Publish needs, upload photos, track pledges, update completion status.
Alumni (Pledger)	Former student, now a city professional, eager to give back.	Browse needs, make pledges, view impact, share on social media.
Viewer (Read-only)	Community member, journalist, or government official.	Monitor school progress and transparency reports without pledging.

6. Functional Requirements

6.1 Authentication & Onboarding

ID	Requirement
FR-01	Admin login via email/password through Firebase Authentication.
FR-02	Alumni self-registration with name, graduation year, and current city.
FR-03	Role-based access control — Admin features hidden from Alumni.

6.2 Needs Management (Admin)

ID	Requirement
FR-04	Admin can create a 'Need Card' with title, description, category, cost estimate, urgency level, and photo.
FR-05	Admin can edit or delete any unpledged need.
FR-06	Admin can mark a need as 'Fulfilled' and upload completion (after) photo.
FR-07	Admin can set a target funding amount and deadline for each need.
FR-08	GenAI Assistant suggests description and cost estimate when Admin types the need title.

6.3 Needs Dashboard (Alumni / All Users)

ID	Requirement
FR-09	Display all active needs as scrollable CardViews sorted by urgency and date.
FR-10	Each card shows: title, category icon, cost estimate, progress bar, and pledged count.
FR-11	Filter needs by category (Furniture, Sanitation, Classroom, Sports, Misc).
FR-12	Search needs by keyword.

6.4 Pledge Module

ID	Requirement
FR-13	Alumni can tap 'Pledge' on any need card to register a simulated commitment.
FR-14	Pledge form captures: pledger name, pledge type (Item / Funds), pledge amount/quantity, and optional message.
FR-15	Progress bar updates in real time (Firebase) as pledges are recorded.
FR-16	Alumni can view and manage their own pledge history.
FR-17	System prevents duplicate pledges from the same user for the same need.

6.5 Impact Gallery

ID	Requirement
FR-18	Fulfilled needs show a Before/After photo comparison (swipeable).
FR-19	Admin can upload before photo at need creation and after photo at need closure.
FR-20	Gallery is publicly viewable; no login required for read-only access.

6.6 Donor Hall of Fame

ID	Requirement
FR-21	Display top pledgers ranked by total simulated pledge value.
FR-22	Each entry shows: name, graduation year, number of pledges, and a badge tier (Bronze / Silver / Gold).
FR-23	Alumnus can share their Hall of Fame card to social media.

6.7 Notifications

ID	Requirement
FR-24	Push notification to all alumni when a new high-urgency need is published.
FR-25	Push notification to all pledgers of a need when it is marked fulfilled.
FR-26	Milestone notification at 25%, 50%, 75%, and 100% funding of any need.

7. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	App should launch in under 3 seconds on mid-range Android (2 GB RAM). Dashboard should load within 2 seconds on a 4G connection.
NFR-02	Scalability	Firebase backend should support up to 10,000 concurrent alumni users without degradation.
NFR-03	Reliability	App uptime of 99.5% (Firebase SLA). Local caching ensures last-known data is visible during brief connectivity loss.
NFR-04	Security	Firebase Authentication with email verification. Firestore security rules restrict write access to authenticated Admin role.
NFR-05	Usability	UI must be intuitive for non-technical headmasters. All primary tasks (publish need, make pledge) completable within 3 taps.
NFR-06	Accessibility	Minimum font size 14sp. Sufficient colour contrast (WCAG AA). Screen-reader support via content descriptions on all interactive elements.
NFR-07	Maintainability	Codebase follows MVVM architecture with clear separation of UI, ViewModel, and Repository layers.
NFR-08	Data Privacy	No sensitive personal data (Aadhaar, bank details) stored. Pledges are simulated commitments only.
NFR-09	Localisation	UI strings externalised in strings.xml; Kannada translation feasible without code changes.
NFR-10	Image Handling	Photos compressed client-side before upload (max 1 MB). Stored in Firebase Cloud Storage with CDN delivery.

8. Feature Prioritisation

Features are classified using the MoSCoW method: Must Have (core MVP), Should Have (important but not launch-blocking), and Could Have (enhances delight — good to have for a polished submission).

8.1 Must Have (MVP — Required for Minimum Viable Product)

- Admin login and role-based access (FR-01, FR-03)
- Create, edit, and delete Need Cards with photo and cost estimate (FR-04, FR-05)
- Needs Dashboard with CardView list and progress bar (FR-09, FR-10)
- Simulated Pledge button and pledge form — no real payment (FR-13, FR-14)
- Real-time progress bar update via Firebase Firestore (FR-15)
- Admin can mark a need as Fulfilled (FR-06)
- Before/After impact photo upload and display (FR-18, FR-19)
- Firebase Authentication for Admin (FR-01)
- Alumni self-registration (FR-02)

8.2 Should Have (Important — Target for Full Submission)

- Donor Hall of Fame with badge tiers (FR-21, FR-22)
- Push notifications for new needs and milestones (FR-24, FR-25, FR-26)
- Filter and search on the Needs Dashboard (FR-11, FR-12)
- Pledge history screen for alumni (FR-16)
- Duplicate pledge prevention (FR-17)
- Offline-tolerant read (last-known data cached)

8.3 Could Have (Good to Have — Elevates Quality)

- GenAI Assistant for auto-generating need descriptions and cost estimates (FR-08) — differentiator feature.
 - Social share card for Hall of Fame entry (FR-23)
 - Urgency badge (Critical / High / Medium) on need cards
 - Animated confetti on 100% pledge milestone
 - Category icons (Material Design) for visual scannability
 - Dark mode support
 - School profile page with history, alumni count, and overall impact statistics
 - CSV export of pledge data for Admin
 - Video proof upload for completed work (alongside photos)
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9. App User Flow

9.1 Admin Flow

- Step 1: Login → School Dashboard → Tap '+' → Create Need Card (Title, Category, Description, Cost, Photo, Urgency) → Publish.
- Step 2: View incoming pledges on each Need Card → progress bar auto-updates.
- Step 3: When need is fulfilled → Upload 'After' photo → Mark as Fulfilled → Card moves to Impact Gallery.
- Step 4: GenAI button on create screen → AI suggests description and cost estimate based on need title.

9.2 Alumni Flow

- Step 1: Register / Login → Land on Needs Dashboard → Browse / Filter needs by category or urgency.
 - Step 2: Tap any Need Card → View detail (description, photos, progress bar, pledger list).
 - Step 3: Tap 'Pledge' → Fill pledge form (type, amount, message) → Confirm → Progress bar updates instantly.
 - Step 4: Receive push notification when pledged need is fulfilled → View before/after photo.
 - Step 5: Check Hall of Fame → See own ranking → Share card to WhatsApp/Instagram.
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10. Technical Architecture & Implementation

10.1 Technology Stack

Component	Technology / Tool
Platform	Android (Kotlin, MVVM Architecture, Jetpack libraries)
UI Components	CardView, RecyclerView, ProgressBar, ConstraintLayout, Material Design 3
Authentication	Firebase Authentication (Email/Password + Google Sign-In optional)
Database	Firebase Firestore (NoSQL, real-time listeners)
Media Storage	Firebase Cloud Storage (compressed JPGs, max 1 MB)
Image Picker	Android ImagePicker / ActivityResultContracts.GetContent
Push Notifications	Firebase Cloud Messaging (FCM)
GenAI Integration	Gemini API (via Android Generative AI SDK) — need description assistant
Dependency Injection	Hilt
Image Loading	Glide or Coil
Version Control	Git / GitHub

10.2 Firebase Data Model (Firestore Collections)

- schools/{schoolId} — name, address, headmasterName, alumniCount, createdAt
 - needs/{needId} — schoolId, title, category, description, costEstimate, urgency, targetAmount, pledgedAmount, status (active/fulfilled), beforePhotoUrl, afterPhotoUrl, createdAt, fulfilledAt
 - pledges/{pledgeId} — needId, alumniId, pledgeType (item/funds), amount, message, timestamp
 - users/{userId} — displayName, email, role (admin/alumni), graduationYear, city, totalPledgeValue, badgeTier
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11. Impact Goals

Community Ownership: Empowering citizens to actively participate in maintaining and improving local public schools — shifting the mental model from 'government responsibility' to 'shared community asset.'

Education Quality: Ensuring infrastructure gaps (broken desks, unusable toilets, peeling classrooms) do not disrupt the learning environment for children.

Transparency & Trust: Building verifiable, photo-evidenced proof of every repair so donors and the public can see their contribution materialise.

Alumni Re-engagement: Reigniting a sense of pride and belonging among former students, creating a virtuous cycle of giving back.

Digital Literacy: Gently introducing headmasters in rural India to digital tools for school management and community communication.

12. Success Criteria

The following criteria define a successful submission for this project:

- ✓ **Needs Dashboard Editability:** Admin can add, edit, delete, and reorder needs without any code changes. All updates are reflected in real time for all logged-in alumni.
- ✓ **Pledge Simulation:** The Pledge button successfully records a commitment in Firebase and visually increments the progress bar — no real financial transaction occurs.
- ✓ **Professional UI:** The interface uses consistent Material Design 3 components, a school-green colour theme, and is inspiring and trust-building in its visual design.
- ✓ **Real-time Updates:** Progress bars and pledge counts update within 2 seconds of a pledge being made, without requiring a manual page refresh.
- ✓ **Impact Proof:** At least one fulfilled need demonstrates a before/after photo pair visible in the Impact Gallery.
- ✓ **GenAI Integration:** The GenAI assistant successfully populates a need description suggestion based on a typed title.
- ✓ **Role Security:** Alumni cannot access Admin-only screens; Firestore security rules prevent unauthorised writes.